The geometric thickness of K_{14} is 4

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31 October 2026

Abstract

No straight-line drawing of K_{14} has its edges split into three crossing-free layers, so the geometric thickness of K_{14} is 4. Together with the upper bound of Dillencourt, Eppstein and Hirschberg this completes the values of $\bar{\theta}(K_n)$ for $n \le 16$. The lower bound is a SAT computation over signotopes with a hull-size case split and an Euler-type edge bound; two independently written encoders give UNSAT, and every leaf LRAT proof was checked with `cake_lpr`.

1 Introduction

Dillencourt, Eppstein and Hirschberg [1] bounded $\bar{\theta}(K_n)$ and left K_{13} and K_{14} open. Seamons showed $\bar{\theta}(K_{13}) = 3$ [2]. This note settles K_{14}.

2 Statement

The geometric thickness $\bar{\theta}(G)$ is the least t such that G has a straight-line drawing whose edges split into t crossing-free classes [1].

Theorem 1. $\bar{\theta}(K_{14}) = 4$. *Equivalently, no straight-line drawing of K_{14} has its edges split into three crossing-free layers.*

- Upper bound: $\bar{\theta}(K_n) \le \lceil n/4 \rceil$ [1, Thm. 2.1]. An explicit integer drawing with four layers: the three-layer K_{13} drawing of [2] plus the point $(1000003, 777781)$, whose 13 edges form the fourth layer.
- Lower bound: new, by the computation below.

Corollary 2. *With [1] and [2], $\bar{\theta}(K_n)$ is 1 for $n \le 4$, 2 for $5 \le n \le 8$, 3 for $9 \le n \le 13$ and 4 for $14 \le n \le 16$.*

3 Encoding

Points $1, \dots, n$ in general position, sorted by x; $X(a, b, c)$ is the orientation of $a < b < c$.

(S) Signotope axioms on every 4-set [4, 3].

(K) $Y(pq, rs) \leftrightarrow [o(pqr) \neq o(pqs)] \wedge [o(rsp) \neq o(rsq)]$; crossing edges never share a layer.

(N) Normal form $X(1, a, b) = +$ for $1 < a < b$ [3]; layer value-precedence; (authors' encoder) the reflection predicate of [3].

(E) With h hull vertices and U the set of uncrossed edges, $|L_k \cup U| + h \le 3n - 3$ for each layer, and $(t-1)|U| + th \le t(3n-3) - \binom{n}{2}$.

Lemma 3. *Every straight-line drawing of K_n with t crossing-free layers has an image satisfying (S), (K), (N), (E).*

Proof outline.

- no collinear triple; rotation to distinct x; signotope axioms for x-sorted sets;
- projective map sending a hull vertex first and the rest in angular order; mirror if needed;
- triangulation edge count $3n - 3 - h$; uncrossed edges can join any layer; summing over layers.

□

4 Case split and certificates

- The normal form makes 1, 2, n hull vertices. Cases by the other hull vertices $S \subseteq \{3, \dots, 13\}$: $|S| \le 2$ exactly (67 cases) and $|S| \ge 3$ by 165 covers $T \subseteq S$, $|T| = 3$; 232 cases.
- Optional sweep lemmas for $|S| \le 2$; a second run without them is also UNSAT.
- Cube-and-conquer with CaDiCaL 3.0.1: 307 leaves per run. Every leaf LRAT verified by `cake_lpr` [5].
- Independent encoder and cube partition (referee): 232 roots, 352 leaves, all verified.
- Completeness of each tree is checked separately, and every leaf CNF is regenerated and compared by SHA-256.

5 Data and verification

Code, case trees with CNF and LRAT hashes, witnesses and a one-command checker: `https://github.com/LeeMarshall1113/ai-math-results`, archived on Zenodo, doi:10.5281/zenodo.23748157 (all versions).

- The LRAT proofs (about 56 GB in total) are not archived; `FULL=1 check_all.sh` regenerates and checks them in about 3.5 hours on one thread.
- The encodings' soundness rests on Lemma 3 and the case split, proved by hand; there is no formal proof that the generators implement them.

6 Discussion

Open: a computer-free proof that $\bar{\theta}(K_{14}) > 3$, and the value of $\bar{\theta}(K_{17})$.

AI use

This work was carried out with AI assistance (Claude, Anthropic), directed by the author. The computations were independently re-implemented and checked, and the result was attacked by adversarial AI referees whose reports are in the repository. It has not been peer reviewed. The author is responsible for all content.

References

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[3] M. J. H. Heule and M. Scheucher, Happy ending: an empty hexagon in every set of 30 points, in *TACAS 2024*, LNCS 14570, 61–80.

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